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POINCARÉ & LOG-SOBOLEV FOR

THE ORNSTEIN-UHLENBECK
AND LANGEVIN-SCHMOLUCHOWSKI SEMIGROUPS

- $B(t), 0 \leq t < \infty$: Standard Brownian Motion in $\mathbb{R}^n$
- $\mathcal{J}$: Standard Gaussian distribution on $\mathcal{B}(\mathbb{R}^n)$, i.e.,

$$\mathcal{J}(dy) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}|y|^2} dy$$

- $X(t), 0 \leq t < \infty$: Ornstein-Uhlenbeck diffusion process

$$dX(t) = -X(t)dt + \sqrt{2} dB(t), \quad X(0) = \xi;$$

equivalently,

$$X(t) = e^{-t} \left(\xi + \sqrt{2} \int_0^t e^s dB(s) \right)$$

where ξ is a random variable independent of $B(\cdot)$.

- Then

$$\begin{aligned} (P_t f)(x) &\triangleq \mathbb{E}^x f(X(t)) \\ &= \int_{\mathbb{R}^n} f(xe^{-t} + y\sqrt{1-e^{-2t}}) \mathcal{J}(dy) \end{aligned}$$

defines the so-called ORNSTEIN-UHLENBECK Semigroup
for $t \in [0, \infty)$, $f \in C_b(\mathbb{R}^n)$.

This has the following properties.

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• $P_0 = I = \text{identity}$

• $[0, \infty) \ni t \rightarrow P_t f \in L^2(\mathbb{R}^n, \gamma)$ is continuous, for every given $f \in C_b(\mathbb{R}^n)$

• $P_t(P_s f) = P_{t+s} f = P_s(P_t f), \quad \forall s \geq 0, t \geq 0$

• $P_t 1 = 1; \quad P_t f \geq 0$ if $f \geq 0$

• $\|P_t f\|_\infty \leq \|f\|_\infty, \quad \forall t \geq 0, f \in C_b(\mathbb{R}^n).$

PROPOSITION: The linear operator

(1)

$$Lf \triangleq \Delta f - \langle x, \nabla f \rangle$$

is the infinitesimal generator of the OU-semigroup:

(2)

$$\frac{\partial}{\partial t} (P_t f)(x) = L(P_t f)(x) = P_t (Lf)(x)$$

for every $t \geq 0, x \in \mathbb{R}^n, f \in C_b^2(\mathbb{R}^n)$. Also $(Lf)(x) = \lim_{t \downarrow 0} \frac{(P_t f)(x) - f(x)}{t}$.

PROPOSITION: The probability measure γ is the unique invariant measure for the OU-semigroup, i.e.,

(3)

$$\int_{\mathbb{R}^n} (P_t f)(x) \gamma(dx) = \int_{\mathbb{R}^n} f(x) \gamma(dx), \quad \forall f \in C_b(\mathbb{R}^n), \quad \forall t \geq 0$$

and is also ergodic:

(4)

$$\lim_{t \rightarrow \infty} (P_t f)(x) = \int_{\mathbb{R}^n} f(y) \gamma(dy); \quad \forall x \in \mathbb{R}^n, f \in C_b(\mathbb{R}^n).$$

Remark: The invariance property (3) is tantamount to

(5)'

$$\int_{\mathbb{R}^n} (Lf)(y) \cdot \gamma(dy) = 0, \quad \forall f \in C_b^2(\mathbb{R}^n)$$

In fact, we have the stronger property

(5)

$$\int_{\mathbb{R}^n} g(y) \cdot (Lf)(y) \cdot \gamma(dy) = \int_{\mathbb{R}^n} f(y) \cdot (Lg)(y) \cdot \gamma(dy) = - \int_{\mathbb{R}^n} \langle \nabla f(y), \nabla g(y) \rangle \gamma(dy)$$

for every f, g in $C_b^2(\mathbb{R}^n)$.

Proof of (5): This is a simple integration by parts, noting that the probability density

of the standard Gaussian measure γ , satisfies the property

$$\phi(y) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}|y|^2}$$

(6)

$$\nabla \phi(y) = -y \phi(y), \quad y \in \mathbb{R}^n.$$

Proof of (2): We have clearly that

$$\frac{\partial}{\partial t} (P_t f)(x) = \int_{\mathbb{R}^n} \left(-x e^{-t} + y \frac{e^{-2t}}{\sqrt{1-e^{-2t}}} \right) \nabla f(x e^{-t} + y \sqrt{1-e^{-2t}}) \gamma(dy)$$

is the sum of the two terms

$$(7) \quad -x e^{-t} \int_{\mathbb{R}^n} \nabla f(x e^{-t} + y \sqrt{1-e^{-2t}}) \gamma(dy) = -x e^{-t} (P_t(\nabla f))(x) = -x \cdot \nabla (P_t f)(x)$$

and

$$\frac{e^{-2t}}{\sqrt{1-e^{-2t}}} \int_{\mathbb{R}^n} y \cdot \nabla f(x e^{-t} + y \sqrt{1-e^{-2t}}) \phi(y) dy = \Delta (P_t f)(x).$$

For this last identity we integrate by parts, recalling (6).

3.a

Remark: (i) The equality (5) is equivalent to

$$\int_{\mathbb{R}^n} g \cdot P_t f \cdot dy = \int_{\mathbb{R}^n} f \cdot P_t g \cdot dy, \quad \forall t \geq 0.$$

Then with $X(0) \sim \mathcal{P}$ (starting the diffusion at its invariant distribution) and selecting $f = \mathbb{1}_A$, $g = \mathbb{1}_B$ we obtain reversibility:

$$\mathbb{P}[X(t) \in A, X(0) \in B] = \mathbb{P}[X(0) \in A, X(t) \in B].$$

(ii) We introduce also the DIRICHLET Form

$$\mathcal{E}(f, g) \triangleq \int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle dy = \int_{\mathbb{R}^n} f (-Lg) dy = \int_{\mathbb{R}^n} g (-Lf) dy$$

and the DIRICHLET Energy

$$\mathcal{E}(f, f) \triangleq \int_{\mathbb{R}^n} |\nabla f|^2 dy = \int_{\mathbb{R}^n} f (-Lf) dy.$$

In particular, $-L$ is a positive operator.

Remark: As we noticed in (7),

$$\nabla(P_t f)(x) = e^{-t} P_t(\nabla f)(x)$$

holds for $t \geq 0$, $x \in \mathbb{R}^n$ and $f \in C_b^2(\mathbb{R}^n)$. In particular, we have also the Commutation property

$$(8) \quad \|\nabla(P_t f)(x)\| \leq e^{-t} P_t(\|\nabla f\|)(x),$$

for all norms $\|\cdot\|$ on $\mathbb{R}^n$.

We define now, for $f \in C_b^2(\mathbb{R}^n)$, its variance

$$(9) \quad \text{Var}_J(f) \triangleq \int_{\mathbb{R}^n} f^2 dy - \left(\int_{\mathbb{R}^n} f dy \right)^2$$

relative to the standard Gaussian J .

THEOREM: POINCARÉ INEQUALITY. The quantity (9) satisfies

$$(10) \quad \text{Var}_J(f) \leq \int_{\mathbb{R}^n} |\nabla f|^2 dy = \mathcal{E}(f, f)$$

with equality if $\nabla f = C \in \mathbb{R}^n$.

$$(10)' \quad \text{Proof: We note } P_0 f = f, P_\infty f = \int_{\mathbb{R}^n} f dy, \text{ introduce } V_t^f \triangleq \int_{\mathbb{R}^n} (P_t f)^2 dy, \text{ and observe}$$

$$\text{Var}_J(f) = V^f(0) - V^f(\infty) = - \int_0^\infty \frac{d}{dt} V_t^f dt.$$

But from (5) we observe

$$(10)'' \quad \begin{aligned} \frac{d}{dt} V_t^f &= 2 \int_{\mathbb{R}^n} \frac{d}{dt} P_t f \cdot P_t f dy = \\ &= 2 \int_{\mathbb{R}^n} L(P_t f) \cdot P_t f dy = -2 \int_{\mathbb{R}^n} |\nabla(P_t f)|^2 dy = -2 \int_{\mathbb{R}^n} \mathcal{E}(P_t f, P_t f) dy; \end{aligned}$$

invoking now (8), we obtain

$$\begin{aligned} \text{Var}_{\mathcal{P}}(f) &= 2 \int_0^\infty \left(\int_{\mathbb{R}^n} |\nabla(P_t f)|^2 dy \right) dt \leq 2 \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} \left(P_t(|\nabla f|) \right)^2 dy \right) dt \\ &\leq 2 \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} P_t(|\nabla f|^2) dy \right) dt. \end{aligned}$$

But the standard Gaussian distribution is invariant under the action of the semigroup $(P_t)_{t \geq 0}$, therefore the spatial integral of the last display is constant, and

$$\text{Var}_{\mathcal{P}}(f) \leq 2 \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} |\nabla f|^2 dy \right) dt = \int_{\mathbb{R}^n} |\nabla f|^2 dy,$$

as claimed.

We note that all inequalities above, hold as equalities for functions f with constant gradient $\nabla f = C \in \mathbb{R}^n$. $\square$

For a given function $f: \mathbb{R}^n \rightarrow [0, \infty)$ with $\int_{\mathbb{R}^n} f dy > 0$, we define now the relative entropy

$$(11) \quad \text{Ent}_{\mathcal{P}}(f) \triangleq \int_{\mathbb{R}^n} f \log \left(\frac{f}{\int_{\mathbb{R}^n} f dy} \right) dy = \mathbb{E}_{\mathcal{P}}[h(f)] - h(\mathbb{E}_{\mathcal{P}}(f)) \geq 0.$$

Here, $\mathbb{E}_{\mathcal{P}}(\cdot)$ denotes integration in $\mathbb{R}^n$ with respect to the standard Gaussian probability measure $\mathcal{P}$, and

$$h(x) \triangleq x \cdot \log x, \quad x > 0$$

is a convex function.

THEOREM: LOG-SOBOLEV INEQUALITY. In the above context,

$$(12) \quad \text{Ent}_{\mathcal{P}}(f) \leq \frac{1}{2} \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f} dy.$$

And again, this holds as equality when $\nabla f = C$ is a constant in $\mathbb{R}^n$.

Remark: This inequality appears often with f replaced by f^2 , in which case it reads

(12)'

$$\frac{1}{2} \text{Ent}_{\mathcal{J}}(f^2) \leq \int_{\mathbb{R}^n} |\nabla f|^2 dy = \mathcal{E}(f, f).$$

Proof: We mimic the previous argument: we introduce

$$E^f(t) \triangleq \int_{\mathbb{R}^n} P_t f \cdot \log P_t f \, dy, \text{ recall } P_0 f = f, P_\infty f = \int_{\mathbb{R}^n} f \, dy, \text{ and note}$$

$$\begin{aligned} \text{Ent}_{\mathcal{J}}(f) &= \int_{\mathbb{R}^n} f \log f \, dy - \int_{\mathbb{R}^n} f \, dy \cdot \log \left(\int_{\mathbb{R}^n} f \, dy \right) = E^f(0) - E^f(\infty) \\ &= - \int_0^\infty \frac{d}{dt} E^f(t) \, dt. \end{aligned}$$

Now from (2), (5') and (5), we get

$$\begin{aligned} \frac{d}{dt} E^f(t) &= \frac{d}{dt} \int_{\mathbb{R}^n} h(P_t f) \, dy = \int_{\mathbb{R}^n} h'(P_t f) \cdot L(P_t f) \, dy \\ &= \int_{\mathbb{R}^n} L(P_t f) \cdot \log(P_t f) \, dy = - \int_{\mathbb{R}^n} \nabla(P_t f) \cdot \nabla(\log P_t f) \, dy \\ &= - \int_{\mathbb{R}^n} \frac{|\nabla(P_t f)|^2}{P_t f} \, dy \end{aligned}$$

where again $h(x) \triangleq x \cdot \log x$.

The quantity

(13)

$$\underline{I_{\mathcal{J}}(f) \triangleq \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f} \, dy}$$

is the FISHER Information of f with respect to the standard GAUSSIAN measure $\mathcal{J}$.

We have derived the identity

$$\frac{d}{dt} E_t^f = - \mathbb{I}_{\mathcal{P}} (P_t f),$$

which gives

$$\begin{aligned} \text{Ent}_{\mathcal{P}}(f) &= E_t^f(0) - E_t^f(\infty) = \int_0^\infty \mathbb{I}_{\mathcal{P}}(P_t f) = \int_0^\infty \left(\int_{\mathbb{R}^n} \frac{|D(P_t f)|^2}{P_t f} dy \right) dt \\ &\leq \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} \frac{(P_t(|Df|))^2}{P_t f} dy \right) dt, \end{aligned}$$

by virtue of (8).

But from CAUCHY-SCHWARZ we get

$$P_t(|Df|) = P_t \left(\frac{|Df|}{\sqrt{f}} \sqrt{f} \right) \leq \sqrt{P_t \left(\frac{|Df|^2}{f} \right) \cdot P_t(f)},$$

thus

$$\frac{P_t(|Df|)^2}{P_t(f)} \leq P_t \left(\frac{|Df|^2}{f} \right). \quad \text{Substituting, we get}$$

$$\text{Ent}_{\mathcal{P}}(f) \leq \int_0^\infty e^{-2t} \left(\int_{\mathbb{R}^n} P_t \left(\frac{|Df|^2}{f} \right) dy \right) dt = \frac{1}{2} \int_{\mathbb{R}^n} \frac{|Df|^2}{f} dy$$

by the invariance of the standard Gaussian measure γ for the OU-semigroup. $\square$

The POINCARÉ and LOG-SOBOLEV inequalities have the following interesting corollaries:

$$(14) \quad \text{Var}_{\mathcal{P}}(P_t f) \leq e^{-2t} \cdot \text{Var}_{\mathcal{P}}(f), \quad \text{Ent}_{\mathcal{P}}(P_t f) \leq e^{-2t} \cdot \text{Ent}_{\mathcal{P}}(f)$$

for all $t \geq 0$ and $f \in L^2(\gamma)$ (resp., $f \in L \log L(\gamma)$).

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$$\left(V_{\pm}^{\pm f}(0) - V_{\pm}^{\pm f}(\infty) = \int_{\mathbb{R}^n} (P_0(P_{\pm}f))^2 dy - \int_{\mathbb{R}^n} (P_{\infty}(P_{\pm}f))^2 dy \right)$$

thus Indeed, from (10)'', (10)', (10) we obtain $\text{Var}_{\mathcal{P}}(P_{\pm}f) = V_{\pm}^{\pm f} = V_{\pm}^{\pm f}(0) - V_{\pm}^{\pm f}(\infty)$,

$$\frac{d}{dt} \text{Var}_{\mathcal{P}}(P_{\pm}f) = -2 \int_{\mathbb{R}^n} |\nabla(P_{\pm}f)|^2 dy \leq -2 \cdot \text{Var}_{\mathcal{P}}(P_{\pm}f),$$

and the first inequality in (14) follows from the GRONWALL inequality. Likewise, from (12) - (13) we get

$$\frac{d}{dt} \text{Ent}_{\mathcal{P}}(P_{\pm}f) = - \int_{\mathbb{R}^n} \frac{|\nabla(P_{\pm}f)|^2}{P_{\pm}f} dy \leq -2 \cdot \text{Ent}_{\mathcal{P}}(P_{\pm}f),$$

from which the second inequality in (14) follows. $\square$

Remark: In addition, we have the TALAGRAND Inequality discussed in class (cf. Exercise 6 in HWK 10):

$$W_2(\mu, \gamma) \leq \sqrt{2 \cdot H(\mu|\gamma)} \quad \text{for } \mu \in \mathcal{P}_2(\mathbb{R}^n).$$

↑
Quadratic
Wasserstein
Distance

↑
K-L Relative Entropy

$$E^{\mu} \log\left(\frac{d\mu}{d\gamma}\right) = \int_{\mathbb{R}^n} f \log f dy, \quad f \triangleq \frac{d\mu}{d\gamma}$$

LS-DIFFUSIONS

We fix now a potential, meaning a function $\Psi: \mathbb{R}^n \rightarrow [0, \infty)$ of class $C^2(\mathbb{R}^n)$, and consider the corresponding LANGEVIN-SCHMOLUCHOWSKI diffusion

$$dX(t) = -\nabla \Psi(X(t)) dt + \sqrt{2} dB(t)$$

with $B(\cdot)$ standard Brownian Motion in $\mathbb{R}^n$. Assuming

$$Z = \int_{\mathbb{R}^n} e^{-\Psi(x)} dx < \infty,$$

this has

$$\mathcal{P}_\Psi(dy) = \frac{1}{Z} e^{-\Psi(y)} dy$$

as its invariant distribution (just as the standard GAUSSIAN is the — of the OU-process when $\Psi(x) = |x|^2/2$).

LS process is

$$Lf = \Delta f - \langle \nabla \Psi, \nabla f \rangle,$$

(15)

again by analogy with (1). And it is checked easily, that we have again the properties

(16)'

$$\int_{\mathbb{R}^n} Lf \cdot d\mathcal{P}_\Psi = 0$$

(16)

$$\int_{\mathbb{R}^n} g \cdot Lf \cdot d\mathcal{P}_\Psi = \int_{\mathbb{R}^n} f \cdot Lg \cdot d\mathcal{P}_\Psi = - \int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle d\mathcal{P}_\Psi$$

as in (5), (5)'.

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We re-write this expression in terms of the positive operator $-L$ and the DIRICHLET Form

$$(16)'' \quad \mathcal{E}_{\psi}^{\ell}(f, g) \triangleq \int_{\mathbb{R}^n} \langle \nabla f, \nabla g \rangle d\gamma_{\psi} = \int_{\mathbb{R}^n} g(-L f) d\gamma_{\psi} = \int_{\mathbb{R}^n} f(-L g) d\gamma_{\psi}.$$

We recall now the expression of Exercise 2 in HWK #10, which involves path integration on the space of continuous functions under WIENER Measure, and note

that we do NOT have now an expression for the semigroup

$$\mathbb{R}^n \ni x \mapsto (P_t f)(x) = \mathbb{E}^x(f(X|t)) \quad , \quad t \geq 0$$

as explicit and crisp as in the OU case.

So, in order to study this semigroup and obtain for it functional inequalities of the LOG-SOBOLEV or POINCARÉ type, we need to develop some new machinery.

To this end we introduce, with BAKRY-ÉMERY (1985), the Bilinear Form

$$(17) \quad \Gamma(f, g) \triangleq \frac{1}{2} [L(fg) - f \cdot L(g) - g \cdot L(f)] = \langle \nabla f, \nabla g \rangle$$

defined for functions f, g in a suitable set which is dense in $L^2(\mathcal{X}_\psi)$

$A \subset L^2(\mathcal{X}_\psi)$

Exercise

(in the OU case, we can take $A = C_c^\infty(\mathbb{R}^n)$). We denote

$$(17)' \quad \Gamma(f) \triangleq \Gamma(f, f) = \frac{1}{2} L(f^2) - f \cdot L(f) = |\nabla f|^2$$

$$(18) \quad \Gamma_2(f) \triangleq \frac{1}{2} L(\Gamma(f)) - \Gamma(f, Lf) = \|\mathcal{D}^2 f\|_{\text{HS}}^2 + \langle \nabla f, \mathcal{D}^2 \Psi \cdot \nabla f \rangle$$

Exercise

Here and below, matrix

$f \in C^2(\mathbb{R}^n)$,

$\mathcal{D}^2 f$
 $\left(D_{ij}^2 f \right)_{1 \leq i, j \leq n}$

denotes the HESSIAN of a function and

$$\|D^2 f\|_{HS} \triangleq \sqrt{\sum_i \sum_j (D_{ij}^2 f)^2}$$

is the HILBERT-SCHMIDT norm of this matrix.

PROPOSITION: BAKRY-ÉMERY (1985)

Suppose the potential Ψ satisfies $D^2 \Psi(x) \geq \rho \cdot \text{Id}$

(19) for all $x \in \mathbb{R}^n$; i.e., $\langle y, D^2 \Psi(x) y \rangle \geq \rho |y|^2, \forall y \in \mathbb{R}^n$

(20) for some $\rho \in \mathbb{R}$.
Then $\Gamma_2(f) \geq \rho \cdot \Gamma(f), \forall f \in \mathcal{A}$.

THEOREM: Suppose the condition (19) is satisfied for some $\rho \in (0, \infty)$. Then we have the POINCARÉ and LOG-SOBOLEV inequalities

(21) $\text{Var}_{\mathcal{H}_\Psi}(f) \leq \frac{1}{\rho} \int_{\mathbb{R}^n} |\nabla f|^2 d\gamma_\Psi, \quad \text{Ent}_{\mathcal{H}_\Psi}(f) \leq \frac{1}{2\rho} \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f} d\gamma_\Psi$

as well as the exponential decay of the variance and relative entropy

(22) $\text{Var}_{\mathcal{H}_\Psi} \left(\frac{Pf}{t} \right) \leq e^{-\frac{2t}{\rho}} \text{Var}_{\mathcal{H}_\Psi}(f), \quad \text{Ent}_{\mathcal{H}_\Psi} \left(\frac{Pf}{t} \right) \leq e^{-\frac{2t}{\rho}} \text{Ent}_{\mathcal{H}_\Psi}(f)$

for $f \in \mathcal{A}$ (resp., $f > 0, f \in \mathcal{A}$).

Remark: More generally, we define in the spirit of (18) the second-order bilinear form

$$(18)' \quad \underline{\Gamma_2(f, g) = \frac{1}{2} [L(\Gamma(f, g)) - \Gamma(f, Lg) - \Gamma(g, Lf)]}, \quad (f, g) \in \mathcal{A} \times \mathcal{A}.$$

As regards the claim of (20), it amounts to

$$\|D^2 f\|_{HS} + \langle \nabla f, D^2 \Psi \cdot \nabla f \rangle \geq \rho \cdot |\nabla f|^2$$

on account of the "Exercises" in (17), (17)', (18); and is obviously satisfied under the condition (19).

Remark: The adjoint L^* of the generator L in (15) is

$$(15)^* \quad \underline{L^* g = \Delta g + \operatorname{div}(g \cdot \nabla \Psi)}$$

This is clear via integration by parts

$$\int_{\mathbb{R}^n} Lf \cdot g \, d\gamma_\Psi = \int_{\mathbb{R}^n} (\Delta f - \langle \nabla f, \nabla \Psi \rangle) g \, d\gamma_\Psi = \int_{\mathbb{R}^n} f (\Delta g + \operatorname{div}(g \cdot \nabla \Psi)) \, d\gamma_\Psi$$

and leads to the FOKKER-PLANCK-KOLMOGOROV equation

$$\frac{\partial}{\partial t} p_t = \Delta p_t + \operatorname{div} \left(p_t \cdot \nabla \Psi \right), \quad t > 0$$

for the probability density function $p_t(\cdot)$ of $X(t)$.

Remark: Both inequalities in (21) can be written in terms of the DIRICHLET Energy

$$\mathcal{E}_{\psi}(f, f) \triangleq \int_{\mathbb{R}^n} |\nabla f|^2 d\gamma_{\psi}$$

as

$$(21)' \quad \text{Var}_{\gamma_{\psi}}(f) \leq \frac{1}{\rho} \mathcal{E}_{\psi}(f, f), \quad \text{Ent}_{\gamma_{\psi}}(f) \leq \frac{2}{\rho} \mathcal{E}_{\psi}(f, f).$$

Remark: Once again, with the KL Relative Entropy

$$H(\mu|\gamma_{\psi}) = \mathbb{E}^{\mu} \log\left(\frac{d\mu}{d\gamma_{\psi}}\right) = \int_{\mathbb{R}^n} f \log f d\gamma_{\psi}, \quad f \triangleq \frac{d\mu}{d\gamma_{\psi}}$$

and the FISHER Information

$$I_{\gamma_{\psi}}(\mu) \triangleq \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f} d\gamma_{\psi},$$

we have the OTTO-VILLANI
(2000) HWI Inequality

$$(23) \quad H(\mu|\gamma_{\psi}) - H(\nu|\gamma_{\psi}) \leq W_2(\mu, \nu) \sqrt{I_{\gamma_{\psi}}(\mu)} - \frac{\rho}{2} W_2^2(\mu, \nu)$$

for any two probability measures μ, ν in $\mathcal{P}_2(\mathbb{R}^n)$. This implies the TALAGRAND Inequality

$$(24) \quad W_2(\mu, \gamma_{\psi}) \leq \sqrt{\frac{2}{\rho} \cdot H(\mu|\gamma_{\psi})}$$

and the Modified LOG-SOBOLEV

$$(25) \quad H(\mu|\gamma_{\psi}) \leq \frac{1}{2\rho} I_{\gamma_{\psi}}(\mu).$$

Remark: Once again, the LOG-SOBOLEV inequality in (21) can be cast, always under the assumption (19), in the equivalent form

$$(21)' \quad \frac{p}{2} \text{Ent}_{\mathcal{H}_\psi}(f^2) \leq \int_{\mathbb{R}^n} |\nabla f|^2 d\mathcal{H}_\psi.$$

GROSS (1975) proved that this inequality implies (in fact, is equivalent to) the following property of hypercontractivity.

DEFINITION: The LS-semigroup $(P_t f)_{t \geq 0}$ is called hypercontractive if, for some (then all) $1 < p < q < \infty$

$$\|P_t f\|_q \leq \|f\|_p, \quad \forall f \in L^p(\mathcal{H}_\psi)$$

holds for all $t > 0$ large enough, namely, $t > \frac{1}{2p} \log\left(\frac{q-1}{p-1}\right)$.

An interesting proof of this result in the special case of the OU-semigroup, based on Stochastic Analysis, appears on pp. 367-369 of IKEDA-WATANABE (1988).

CONCENTRATION OF MEASURE

It turns out that a LOG-SOBOLEV inequality of the form

$$(26) \quad \text{Ent}_\mu(f^2) \leq 2C \int_{\mathbb{R}^n} |\nabla f|^2 d\mu$$

for every sufficiently smooth function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and relative to a given probability measure μ on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, implies the concentration of measure inequality

$$(27) \quad \mu\left(\left\{F - \int_{\mathbb{R}^n} F d\mu \geq r\right\}\right) \leq e^{-\frac{r^2}{2C}}, \quad r > 0$$

for every integrable and Lip(1)-function $F: \mathbb{R}^n \rightarrow \mathbb{R}$.

(Here, we can replace $\mathbb{R}^n$ by an arbitrary metric space equipped with the BOREL σ -algebra generated by its open sets.)

Proof: Consider an integrable, LIPSCHITZ-continuous function

$F: \mathbb{R}^n \rightarrow \mathbb{R}$ with $\int_{\mathbb{R}^n} F d\mu = 0$. Assuming, as we may, that

F is smooth enough, we may take $|\nabla F| \leq \|F\|_{\text{Lip}}$ everywhere.

We apply then (26) to the function

$$(28) \quad f = e^{\frac{\lambda}{2} F - \frac{\lambda^2}{4} C \|F\|_{\text{Lip}}^2}$$

for $\lambda \in \mathbb{R}$, to obtain

$$\begin{aligned} \int_{\mathbb{R}^n} |\nabla f|^2 d\mu &= \frac{\lambda^2}{4} \int_{\mathbb{R}^n} |\nabla F|^2 e^{\frac{\lambda}{2} F - \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2} d\mu \\ &\leq \frac{\lambda^2}{4} \|F\|_{\text{Lip}}^2 \int_{\mathbb{R}^n} e^{\frac{\lambda}{2} F - \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2} d\mu. \end{aligned}$$

With

$$(29) \quad K(\lambda) \triangleq \int_{\mathbb{R}^n} e^{\lambda F - \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2} d\mu$$

and the choice (28), the LSI inequality (26) gives

$$\begin{aligned} \int_{\mathbb{R}^n} \left(\lambda F - \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2 \right) e^{\lambda F - \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2} d\mu &= K(\lambda) \cdot \log K(\lambda) \\ &\leq \frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2 \cdot K(\lambda). \end{aligned}$$

But this amounts to

$$(30) \quad \lambda K'(\lambda) \leq K(\lambda) \log K(\lambda), \quad \lambda \in \mathbb{R}.$$

Now with $H(\lambda) \triangleq \frac{1}{\lambda} \log K(\lambda)$, $\lambda \in \mathbb{R}$ (thus, by L'HOSPITAL,

$$H(0) = \frac{K'(0)}{K(0)} = \int_{\mathbb{R}} F d\mu = 0), \text{ the inequality (30) amounts to}$$

$H'(\lambda) \leq 0$ for $\lambda \in \mathbb{R}$. Thus $H(\cdot)$ is non-increasing, which means

$K(\lambda) \leq 1$, $\forall \lambda \geq 0$. In other words

$$(31) \quad \int_{\mathbb{R}^n} e^{\lambda F} d\mu \leq e^{\frac{C}{2} \lambda^2 \|F\|_{\text{Lip}}^2}$$

from (29), for all $\lambda \geq 0$. And replacing F by a smooth convolution, we see that (31) holds for every Lipschitz function with zero mean.

If, in addition, $\|F\|_{\text{Lip}} \leq 1$, then

$$\mu(\{F \geq r\}) \leq e^{-\lambda r} \int_{\mathbb{R}^n} e^{\lambda F} d\mu \leq e^{C\lambda^2/2 - \lambda r}, \quad r \geq 0.$$

Optimizing over $\lambda \geq 0$, we obtain $\mu(\{F \geq r\}) \leq e^{-r^2/2C}$,

and the claim (27) follows readily. $\square$